📈 Loan Collectability & Recovery Analytics Toolbox

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Target Stakeholders: Recovery Operations & Strategy (1st & 2nd Line of Defense)

A Streamlit-based interactive analytics dashboard that can be used for modeling **loan collectability**, **expected recovery value**, and **portfolio-level cash-flow projections**. This app allows analysts to upload real data or generate synthetic demo portfolios, visualize recovery potential, view goals, expected recoveries, and export results for further analysis.

🚀 Features

🔷 1. Synthetic or Uploaded Data

- Upload your own **CSV or Excel** file
  OR
- Generate a synthetic portfolio with adjustable account count

🔷 2. Recovery Probability & NPV Modeling

The app computes:

- Recovery probability (based on delinquency & strategy)
- Expected Net Present Value (NPV)
- Estimated recovery month
- Delinquency bucket classification

🔷 3. Interactive Visualizations

Includes:

- **Recovery Map (Bubble Chart)** — NPV vs. recency with trendline
- **12-Month Projected Cash Flow**
- **Bucket Concentration Bar Chart**

- **Goal Progress Gauge**

🔷 4. Account Ledger & Export

- Sortable, formatted account table
- Export full results to CSV
- Downloadable Excel template with sample data

🔷 5. Priority Action Items

Automatically identifies the **Top 5 accounts** with highest expected recovery value.

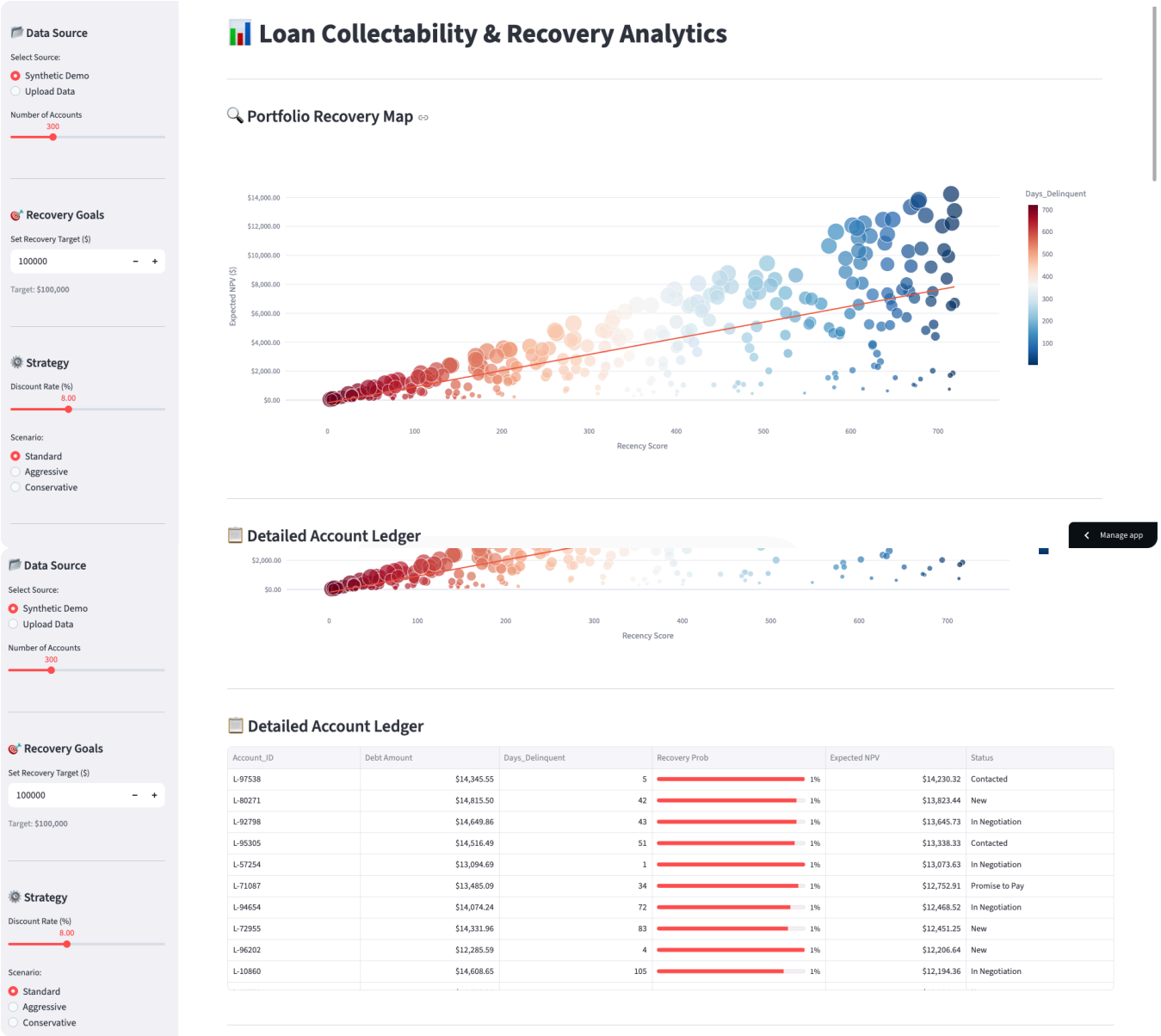
📦 Installation

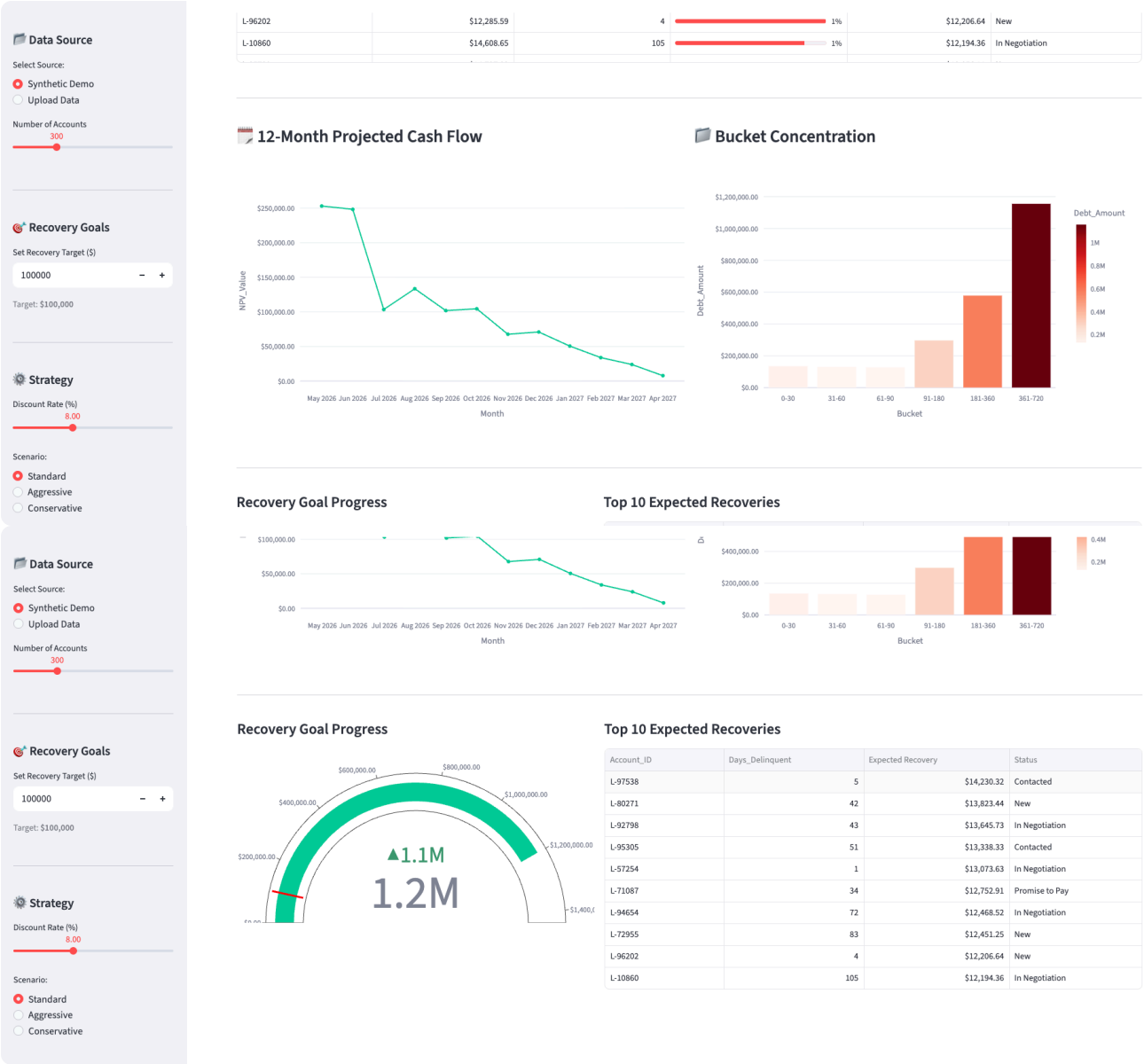
```
# Clone the repository
git clone https://github.com/your-repo/loan-collectability-toolbox.git
cd loan-collectability-toolbox

# Install dependencies
pip install -r requirements.txt

streamlit
pandas
numpy
plotly
xlsxwriter
```

Screenshots





Account_ID	Days_Delinquent	Expected Recovery	Status
L-97538	5	$14,230.32	Contacted
L-80271	42	$13,823.44	New
L-92798	43	$13,645.73	In Negotiation
L-95305	51	$13,338.33	Contacted
L-57254	1	$13,073.63	In Negotiation
L-71087	34	$12,752.91	Promise to Pay
L-94654	72	$12,468.52	In Negotiation
L-72955	83	$12,451.25	New
L-96202	4	$12,206.64	New
L-10860	105	$12,194.36	In Negotiation